ADVANCING EARTH AND SPACE SCIENCES

Space Weather

RESEARCH ARTICLE
10.1029/2025SW004544

The Influence of 3-D Earth Conductivity, Geoelectric Field Polarization, and Power Grid Topology on GIC Risk

Hannah G. Parry[1], Darcy R. Cordell[1,2], Martyn J. Unsworth[1], Ryan MacMullin[3], and Ian R. Mann[1]

[1]Department of Physics, University of Alberta, Edmonton, AB, Canada, [2]Center for Science, University of Athabasca, Athabasca, AB, Canada, [3]AltaLink LP., Edmonton, AB, Canada

Key Points:

- Phase and polarization of E are influenced by the 3D impedance tensor, such that E_x and E_y may be highly correlated in a specific region
- Truth tables show max and min GIC for different combinations of E field polarization direction and power line configuration at a nodal level
- A real-world example shows increased GIC in central AB due to the combination of a NW-SE E-field polarization and the local network topology

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

H. G. Parry,
hparry@ualberta.ca

Citation:

Parry, H. G., Cordell, D. R., Unsworth, M. J., MacMullin, R., & Mann, I. R. (2025). The influence of 3-D earth conductivity, geoelectric field polarization, and power grid topology on GIC risk. *Space Weather*, *23*, e2025SW004544. https://doi.org/10.1029/2025SW004544

Received 14 MAY 2025
Accepted 13 OCT 2025

Author Contributions:

Conceptualization: Hannah G. Parry, Darcy R. Cordell, Martyn J. Unsworth, Ryan MacMullin, Ian R. Mann
Data curation: Hannah G. Parry, Darcy R. Cordell, Ryan MacMullin
Formal analysis: Hannah G. Parry, Darcy R. Cordell, Ian R. Mann
Funding acquisition: Hannah G. Parry, Darcy R. Cordell, Martyn J. Unsworth, Ryan MacMullin, Ian R. Mann
Investigation: Hannah G. Parry
Methodology: Hannah G. Parry, Darcy R. Cordell, Ian R. Mann
Project administration: Ryan MacMullin, Ian R. Mann
Resources: Martyn J. Unsworth, Ryan MacMullin, Ian R. Mann

Abstract Geomagnetically induced currents (GICs) are naturally occurring electrical currents that flow through the Earth and long conductive infrastructure, such as power lines, during geomagnetic storms. GICs in the electric power grid can cause damage to electric power transformers, system failures, and wide-scale blackouts. Here, the combined effects of the power network's topology and the Earth's underlying conductivity structure on GIC risk are examined using examples from the electric power grid in Alberta, Canada. Our results show that the electric field polarization generated by geomagnetic storms is strongly dependent on the subsurface conductivity structure. Moreover, due to Earth induction effects, the two components of the transverse electric fields can also be highly correlated in specific geological regions. Combined, this creates a preferred electric field directionality, presenting a GIC risk for power grids with specific directional topology. A direct comparison between the geoelectric field and the measured transformer neutral-to-ground (TNG) currents measured near Edmonton, Alberta, on 24-04-2023 is shown. At this location, the two horizontal components of the geoelectric field are strongly correlated, and E_x and E_y show strong temporal and waveform correspondence with the TNG current. Two truth tables illustrate the increased or decreased GIC risk in such cases demonstrating that the GIC in a segment of the electric power network may combine constructively or deconstructively depending on the power network configuration and the relative orientation of the geoelectric field polarization. This is further exemplified by a case study of two real-world network configurations in central Alberta.

Plain Language Summary During geomagnetic storms, geomagnetically induced currents (GICs) can flow through the ground and through power lines. These currents can damage transformers and, in severe cases, cause large-scale power outages. In this study, we examine the combined influence of power network topology, local geology and electric field direction on GIC risk in Alberta, Canada. A preferred direction of the polarized electric field can emerge when using a 3-D description of the underlying conductivity of the Earth. We show in some areas of Alberta, the preferred direction of the electric field polarization is northwest-southeast. At some network connections, these effects can combine constructively or destructively, leading to increased or decreased GIC flowing to ground at those locations. To illustrate this, we constructed two truth tables. We also model and demonstrate this effect using a real-world case from a geomagnetic storm on 24 April 2023.

1. Introduction

Geomagnetic disturbance (GMD) events, characterized by temporal and spatial variation of the magnetic field at the surface of the Earth caused by geomagnetic storms, induce electric fields which can drive geomagnetically induced currents (GICs) in electrically conductive infrastructure such as long-distance electric power lines. Since electric power systems are designed to typically operate at 50–60 Hz AC, the flow of near-DC current in the form of GIC can produce an offset of magnetic flux in the substation transformers resulting in saturation in the half-cycle AC current. This can potentially lead to transformer heating and damage, disruption to power system operation (e.g., Molinski, 2002; Pulkkinen et al., 2017; Schrijver et al., 2014) and, in the worst case, electrical blackouts such as those seen in the Hydro-Quebec electric power system in March 1989 (Bolduc, 2002; Boteler et al., 2019).

The source currents in the ionosphere and magnetosphere at high latitudes create complex and spatio-temporally variable GMD fields during geomagnetic storms with the magnitude of GMD variations generally correlating with latitude; therefore, close to and within the auroral oval where GMD are stronger and more frequent, there is a greater risk from GICs (e.g., Pirjola, 2000). Another important factor which influences the magnitude of the

Software: Hannah G. Parry, Darcy R. Cordell
Supervision: Martyn J. Unsworth, Ian R. Mann
Validation: Hannah G. Parry
Visualization: Hannah G. Parry, Darcy R. Cordell
Writing – original draft: Hannah G. Parry, Darcy R. Cordell
Writing – review & editing: Hannah G. Parry, Darcy R. Cordell, Martyn J. Unsworth, Ryan MacMullin, Ian R. Mann

induced geoelectric field is the underlying conductivity of the Earth. In the past, the ground conductivity has often been approximated as a uniform halfspace or represented as a stack of 1-D layers (e.g., NERC, 2020; Pulkkinen et al., 2012; Trichtchenko et al., 2015). In both cases, the electric field direction remains perpendicular to the magnetic field for each frequency (e.g., Cordell & Unsworth, 2025). Thus, for a 1-D subsurface conductivity model, a multidirectional magnetic field will have an associated geoelectric field with no preferred polarization direction. However, true ground conductivity structures usually varies in 2-D or 3-D and can be estimated in the form of an impedance tensor calculated from the horizontal magnetic and electric fields measured during multi-point magnetotelluric (MT) surveys. The impedance tensor, when convolved with the measured ground-based magnetic field in the frequency domain, can be used to estimate the related horizontal geoelectric field components (e.g., Bedrosian & Love, 2015; Cuttler et al., 2018; McKay & Whaler, 2006). As we will show, the 2-D/3-D variation in the subsurface conductivity can result in a preferred geoelectric field polarization direction even when driven by a multidirectional magnetic field. In this scenario, for every frequency, the components of the geoelectric field may not be orthogonal and direction of the horizontal electric field vector is determined mainly by the alignment of subsurface geological structure of the Earth (e.g., Love et al., 2022). The MT method to estimate the geoelectric field using 3-D impedance tensor is particularly useful for calculating the magnitude and direction the GIC is flowing through a given transmission line in models of the electric power grid network where the underlying ground conductivity structure is 2-D or 3-D (e.g., Cordell et al., 2021, 2024; Marshall et al., 2019; Torta et al., 2017; Zheng et al., 2013). In particular, Cordell et al. (2021) highlight the importance of using empirical MT impedance data sets at locations where the subsurface geology contains a discontinuity in the conductivity structure. Such structures can have a profound effect on the amplitude and polarization of the induced geoelectric field. These prior studies have effectively shown that moving from the traditional impedance, calculated from a 1-D layered Earth model, to measured MT impedance data which incorporate 3-D Earth conductivity characteristics should increase GIC model fidelity and therefore improve the accuracy of GIC risk assessments.

The amplitude and spectra of GICs in an element of the power grid (e.g., in a given transmission line or transformer winding, or at a specific substation neutral) are affected by the magnitude and frequency components of the driving GMD, by the underlying conductivity structure of the Earth, and by the topology and connectivity of the power grid itself. A comprehensive and multidisciplinary approach is required both to gain a complete understanding of the GICs that flow at a network level within a specific complex and integrated electrical power network, and to assess the related GIC risks. In this paper, we investigate the role of the 3-D Earth conductivity in influencing the polarization and phase characteristics of the geoelectric field that drives GIC, specifically in the context of its risk impacts for power line connections of a given topological orientation and directionality. As shown here, for some underlying geological structures, the geoelectric field can be highly polarized and coherent, representing a preferentially high risk for an overlying power grid with a specific directional topology. We investigate this phenomenon and provide real-world examples of local ground conductivity and power line geometry conditions which result in enhanced coupling between the geoelectric field and the electric transmission system, thereby generating an increased risk of damage to the electric power grid as a result of GICs driven by space storms.

2. Data and Methods

2.1. Calculating Local Geoelectric Field

In the context of a model of GICs in the electric power grid, the fluctuating magnetic field during a GMD event induces an electric field which generates an electromotive force (emf) that drives GICs in electric transmission line segments (e.g., Boteler & Pirjola, 2017). To accurately estimate and model the emf in a given power line segment, the geoelectric field acting along that line must be accurately estimated. In place of real-time measurements of the geoelectric field, spatial and temporal variations in the geoelectric field can be estimated using the frequency domain characteristics of the GMD and the measured frequency-domain surface impedance, $Z(\omega)$, derived using MT methods (Kelbert, 2019). Z is a complex-valued 2×2 tensor defined by the frequency-dependent ratio of the horizontal electric and magnetic fields measured during an MT survey. Higher frequency signals attenuate more rapidly into the Earth and are thus sensitive to shallower conductivity structures, whereas lower frequency signals penetrate deeper and are thus sensitive to deeper conductivity structures as well.

In this study, the standard geomagnetic coordinate system is used, where the Earth's surface represents the x–y plane, with the x-axis pointing north, the y-axis pointing east, and the z-axis directed downward. A convolution of the transfer function, $Z(\omega)/\mu_0$, where μ_0 is the permeability of free space, and the magnetic field in the frequency domain, $B(\omega)$, gives the estimated geoelectric field, $E(\omega)$. This relation is written and expressed in the frequency domain where $Z(\omega)$ has units of Ohms, E has units Volts per meter, and B has units of Tesla (e.g., Boteler, 1994; Pirjola, 2002). An inverse Fourier transform is applied for an estimate of the geoelectric field time series (Equation 1).

$$\begin{bmatrix} E_x(t) \\ E_y(t) \end{bmatrix} = \mathcal{F}^{-1}\left\{ \frac{1}{\mu_0} \begin{bmatrix} Z_{xx}(\omega) & Z_{yx}(\omega) \\ Z_{xy}(\omega) & Z_{yy}(\omega) \end{bmatrix} \begin{bmatrix} B_x(\omega) \\ B_y(\omega) \end{bmatrix} \right\} \tag{1}$$

In this study, we utilize two pre-existing data sets to compute the geoelectric field during the course of a moderate geomagnetic disturbance event on 24-04-2023. The first was impedance tensors from long-period MT surveys completed by the University of Alberta (e.g., Cordell et al., 2021; Wang & Unsworth, 2022). The second is 1-s magnetic field data from the Canadian Array for Real-time Investigations of Magnetic Activity (CARISMA; Mann et al., 2008) magnetometer stations at multiple locations in Alberta. Over 500 MT impedance measurements exist at locations across the province of Alberta, Canada, three of which will be used for the detailed investigations reported in this paper. We use the impedance tensors calculated at the ABT500, ABT272, and NAB944MT sites, all of which had a sampling rate of 8 Hz. A linear interpolation in log space was applied to the magnitude and phase of each Z component such that Z and B are defined over the same set of frequencies. We chose to use the available magnetometer data from the Ministik Lake (MSTK; 53.351 N, 112.974 W). We refer to Fort McMurray (MCMU; 56.657 N, 111.210 W) and Fort Chipewyan (FCHP; 58.769 N, 111.106 W) CARISMA stations as they are located in regions with different known underlying 3-D Earth conductivity structures (Figure 1). MSTK and MCMU have a separation of about 385 km, MCMU and FCHP have a separation of about 235 km, and MSTK and FCHP have a separation of about 615 km. We do not use magnetometer data from MCMU and FCHP in the analysis in main paper, but refer to their locations as points of reference to the nearby MT sites. MCMU and FCHP magnetic data is used to calculate E in the Supporting Information S1 of this paper.

2.2. Direct Monitoring GICs

The transformer neutral-to-ground (TNG) current, measured with a Hall effect sensor on the transformer grounding connection at a given transformer substation, provides a direct measurement of the GIC flowing to ground at that location. These sensors take precise measurements of the DC component of the current flowing through a grounded transformer. Seven TNG current monitors at six substations in central and southern Alberta were installed between 2019 and 2021 by AltaLink, one of Alberta's largest transmission line companies. Hall sensor data for GMD events of interest is available to us upon request to our industry partner, AltaLink L.P (e.g., Cordell et al., 2024; Parry et al., 2024). We limit this study to three distinct geological regions in central and northern Alberta; therefore do not show GIC data from substation locations in southern Alberta. At the time of publication, there are no Hall sensors installed in northern Alberta near MCMU or FCHP.

This study examines TNG current data collected at two substations in central Alberta. The substations are connected by a 500 kV line orientated east–west. We refer to the substation on the west end of this power line as Substation 1 and the substation on the east end of this power line as Substation 2. Each substation is well-connected to both the 500 and 240 kV portions of the Alberta electric power network.

3. Overview of the Subsurface Earth Conductivity in Alberta

In Alberta and southwestern British Columbia, Canada, an array of over 500 Earth impedance measurements have been made using the MT method. Alberta is the only jurisdiction in Canada with such a densely sampled long-period MT data set, and this has been previously used for tectonics studies and for oil and gas, mineral, and geothermal exploration (Jones et al., 2014; Nieuwenhuis et al., 2014; Türkoğlu et al., 2009; Unsworth, 2015; Wang & Unsworth, 2022). A more detailed and comprehensive description of the 3-D conductivity structure across the province as determined by the MT method is described in Wang and Unsworth (2022).

Figure 1 shows a map of the crustal conductance (i.e., depth-integrated conductivity) across Alberta from the base of the Western Canada Sedimentary Basin (WCSB) to 50 km depth. This figure provides an overview of the

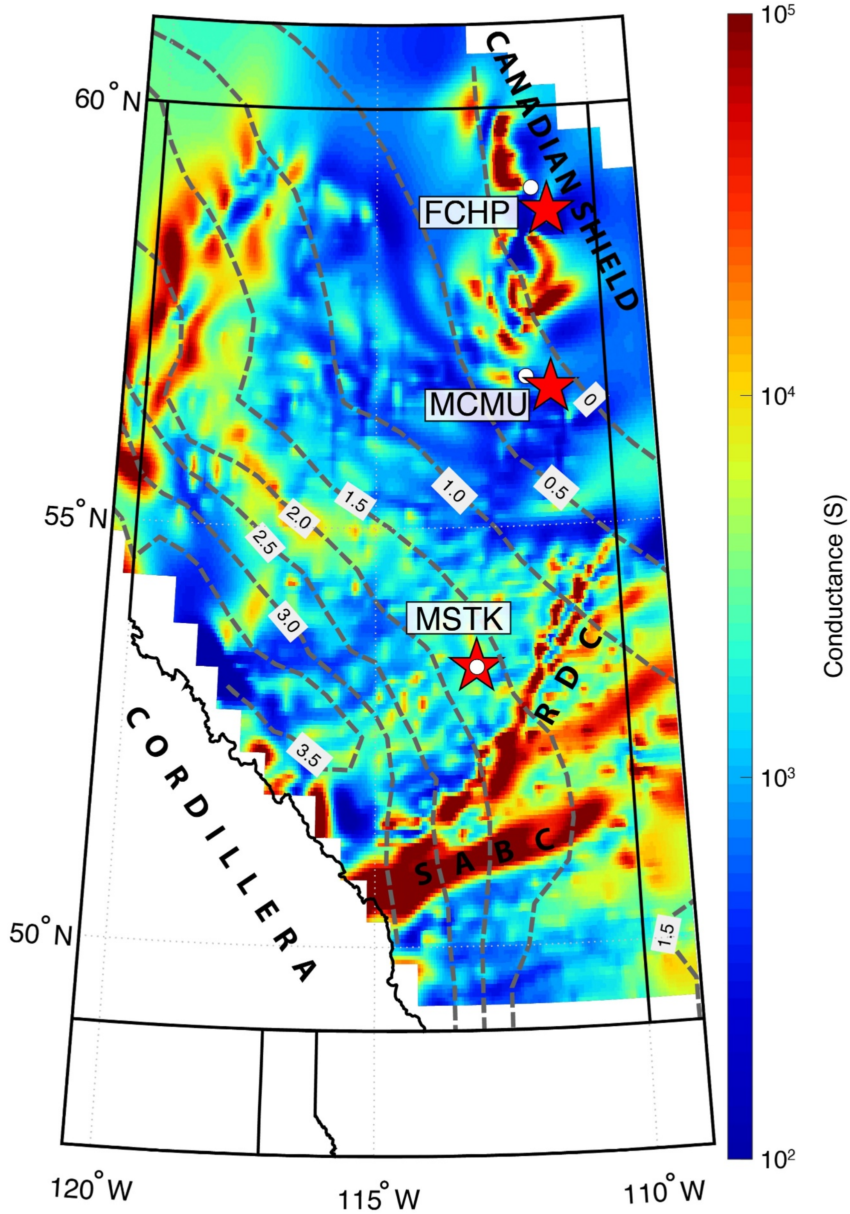

Figure 1. Map of Alberta showing the conductance beneath the province. Conductance is calculated downwards starting from the depth of the base of the Western Canada Sedimentary Basin (WCSB; gray dashed contours in km) to 50 km depth. The Cordillera begins in the southwest where the WCSB contours terminate. The Canadian Shield begins in the northeast at the zero WCSB contour. The Red Deer Conductor (RDC) and the Southern Alberta British Columbia conductor (SABC) are NE-SW trending crustal conductors in south-central Alberta. Red stars denote the three selected CARISMA magnetometer stations at FCHP, MCMU, and MSTK. White circles are the nearest magnetotelluric impedance measurement to each CARISMA magnetometer station.

underlying geology in Alberta and represents the framework within which the effects of the underlying Earth conductance are assessed in this paper. Note that the overlying WCSB is excluded from the conductance calculation because it covers most of the province and is characterized by a wedge of low resistivity (<100 Ωm) sedimentary rocks which tapers to the northeast, with a thickness ranging from 0 to 4 km (Figure 1). Underlying the WCSB, the crust is composed of generally resistive Precambrian (>0.5 Ga) basement rocks and a thick lithosphere extending to depths of >200 km. As the WCSB tapers to the northeast, the high resistivity (>1,000 Ωm) Paleoproterozoic and Archean basement rocks of the Canadian Shield are exposed at the surface in northeastern Alberta (Ross, 2002).

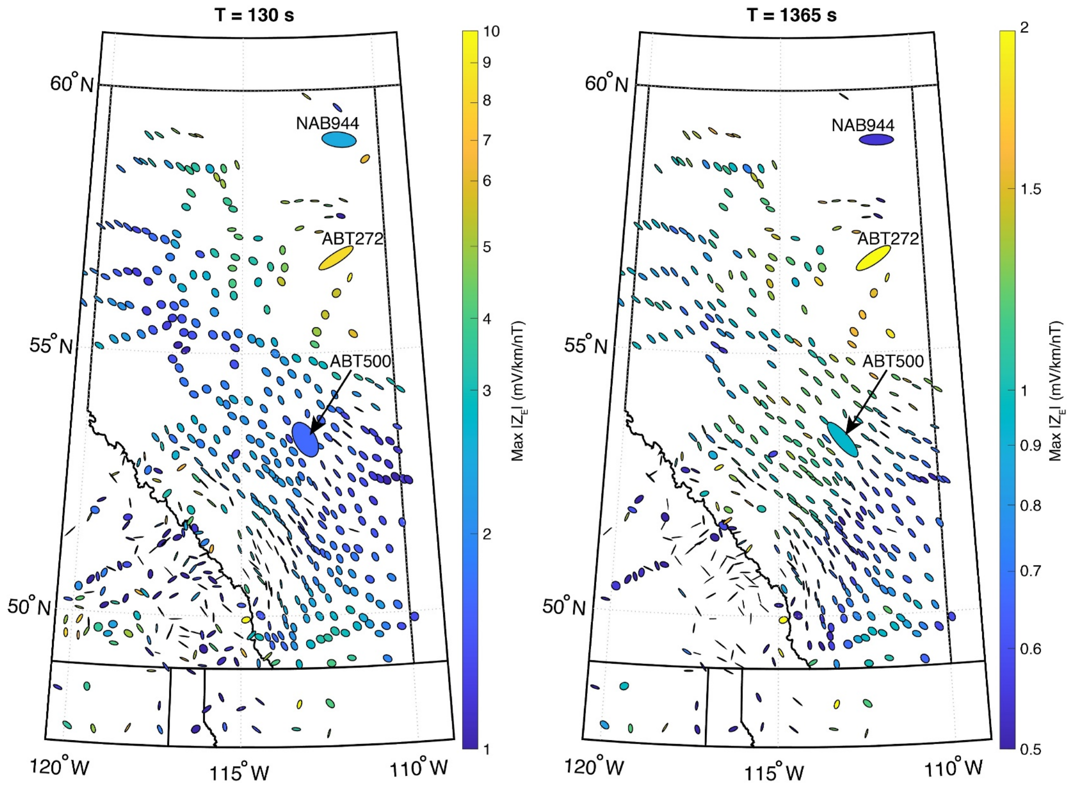

Figure 2. Maximum E-polarized impedance magnitude as a function of the polarization angle at two different periods, 130 s (left) and 1,365 s (right). The three ellipses at the MT sites of interest closest to the MSTK, FCHP and MCMU magnetic stations are enlarged to more clearly indicate the directionality of the impedance close to these three stations.

While the Precambrian basement rocks beneath Alberta are generally resistive, there are several notable high-conductance anomalies trending southwest-northeast beneath the WCSB (Figure 1; see also Wang and Unsworth (2022)). The Red Deer Conductor (RDC) and Southern Alberta British Columbia Conductor (SABC) are associated with the Snowbird Tectonic Zone (STZ) and Vulcan Structure magnetic anomaly, respectively. Both are likely relic features of ancient subduction zone processes during the Paleoproterozoic assembly of Laurentia (i.e., the ancient North American proto-continent) (Nieuwenhuis et al., 2014; Ross & Eaton, 2002; Wang & Unsworth, 2022). To the southwest, the thick WCSB abuts the Canadian Cordillera comprising structurally complex and resistive surface rocks associated with northwest-southeast trending mountain belts and a thin lithosphere (Hanneson & Unsworth, 2023). Since we are limiting this study to the province of Alberta, we do not specifically investigate the influence of impedance on GIC risk in southeastern British Columbia here.

We consider three distinct conductivity regions with measured surface impedances from MT surveys conducted near the three selected CARISMA magnetometer stations. MSTK is located in central Alberta (53.351 N, 112.974 W) and sits above a relatively thick (>1.5 km) WCSB, close to the STZ. MT site ABT500 is nearly co-located with MSTK (53.348 N, 112.969 W), less than 0.5 km away. MCMU is located in northeastern Alberta (56.657 N, 111.210 W) and is underlain by a relatively thin (<0.2 km) WCSB. ABT272 (56.784 N, 111.740 W) is the nearest MT site to MCMU, located 35.3 km away. Finally, FCHP is located in northern Alberta (58.769 N, 111.106 W) atop the outcropping Canadian Shield. The nearest MT site to FCHP is NAB944 59.031 N, 111.440 W, located 34.9 km away. Throughout this paper, we refer to the impedance tensors calculated at the ABT500, ABT272, and NAB944MT sites as MSTK Z, MCMU Z and FCHP Z, respectively. The apparent resistivity and phase as a function of frequency of three distinct conductivity regions, MSTK, MCMU and FCHP, are plotted in the Figure S1 of Supporting Information S1.

In Figure 2, the E-polarized impedance as a function of the polarization axis of the geoelectric field is plotted as an ellipse for two selected frequencies corresponding to periods of 130 and 1,365 s respectively, at MT survey locations across Alberta and southeast British Columbia. This figure illustrates both the frequency-dependent directionality of the MT survey data and the magnitude of the impedance, and was calculated using the mathematical approach described in Berdichevsky and Dmitriev (2008) and which was previously employed for MT surveys in the Unites States by Love et al. (2022). The impedance polarization has 90-degree ambiguity, thus the polarity is either parallel or perpendicular to the semi-major axis. The polarization ellipses are frequency-dependent and the impedance magnitude is typically larger at higher frequencies. Unlike the phase tensor, this E-polarization ellipse includes the effects of galvanic distortion. Whether galvanic distortion should be removed prior to estimating geoelectric fields depends on the spatial scale of the galvanic distorting feature and remains an open question (e.g., Malone-Leigh et al., 2023; Murphy et al., 2021; Utada & Munekane, 2000).

If the subsurface Earth conductivity structure is approximately one-dimensional (i.e., stacked horizontal layers), the ellipses will be circular. If the crustal conductivity structure is approximately two-dimensional (i.e., a straight fault separating two distinct conductive lithologies), then the major axis of the ellipse will typically be aligned parallel/perpendicular to the 2-D conductivity boundary on the conductive/resistive side of the fault, respectively. If the subsurface Earth conductivity structure is three-dimensional, then the ellipse axes can be aligned in various orientations and depending on the spatial scale of the non-uniformity there can also be spatial discontinuity between the ellipses measured at adjacent MT sites.

Figure 2 illustrates both the spatial variations in impedance magnitude, and indicates the geological boundaries of subsurface conductivity structures in Alberta where the orientation of the polarization ellipses is expected to change. In general, a gradual increase in the impedance magnitude can be seen from southwest to northeast across the province. This is a consequence of the thinning of the WCSB wedge and the associated shallowing of the resistive Precambrian basement (Cordell et al., 2021). In general, the polarization semi-major axis of the geoelectric field trends NW-SE consistently across the WCSB, with a stronger and more linear polarization for variations at 1,365 s versus 130 s. This is likely related to the large NE-SW trending crustal conductors (e.g., Cordell et al., 2021). The polarization in northeastern Alberta is more scattered but a lack of spatial coverage makes it difficult to fully establish the spatial trends.

The maximum E-polarized impedance magnitudes in Figure 2 tend to occur in the geologically complex region of the Cordillera. Northeastern Alberta also sees large impedance magnitudes with ABT272 (near MCMU) having values of 8.2 mV/km/nT at 130 s, and 2.1 mV/km/nT at 1,365 s, respectively. This is larger than ABT500 near MSTK which has values of 1.6 mV/km/nT at 130 s, and 0.9 mV/km/nT at 1,365 s, respectively. Despite being located on the Canadian Shield, NAB944 (near FCHP) has a lower magnitude impedance than at MCMU of only 0.5 mV/km/nT at 1,365 s, and has an intermediate impedance value between that at MSTK and MCMU of 2.5 mV/km/nT at 130 s. This highlights that broad regional trends do not always hold, as expected due to smaller-scale geological features.

4. Results

4.1. Characterizing the Geoelectric Field and Resulting GIC Across Alberta

The relationship between the derived horizontal geoelectric field time-series at MSTK and the measured TNG current at a proximal substation during a moderate geomagnetic storm on 24-04-2023 is investigated. A comparison between the two data sets is plotted in Figure 3. During this event, the geoelectric field peak amplitude at MSTK was −0.35 V/km and 0.44 V/km, in the x and y components, respectively. The largest TNG current recorded on a single circuit was 113 A, equivalent to approximately 38 A per phase. Figure 3 shows that the waveforms of the electric fields calculated from impedances and the measured TNG GIC match closely for both small and large temporal scale variations. Thus, the calculated geoelectric field is very clearly closely related to the observed TNG current flowing through the ground connection at the substation. Interestingly, at MSTK station the resulting E_x and E_y components are seen to be highly coherent and in anti-phase. The fact that the TNG GIC at the nearby substation is well-correlated with the electric field, and which is especially clear in the top panel of Figure 3, further indicates not only that the TNG GICs appear to be clearly driven by the derived electric fields but also provides support for the approach of using the impedance tensor together with the observed magnetic field of the GMD to estimate the local electric field.

Interestingly, the largest differences in waveform between the TNG current and the geoelectric field occur at times of the largest geoelectric field variations, between approximately 09:20–10:00 UT. This likely indicates the importance of the frequency and the spatial scale of the GMD for driving GIC in the power network. For example,

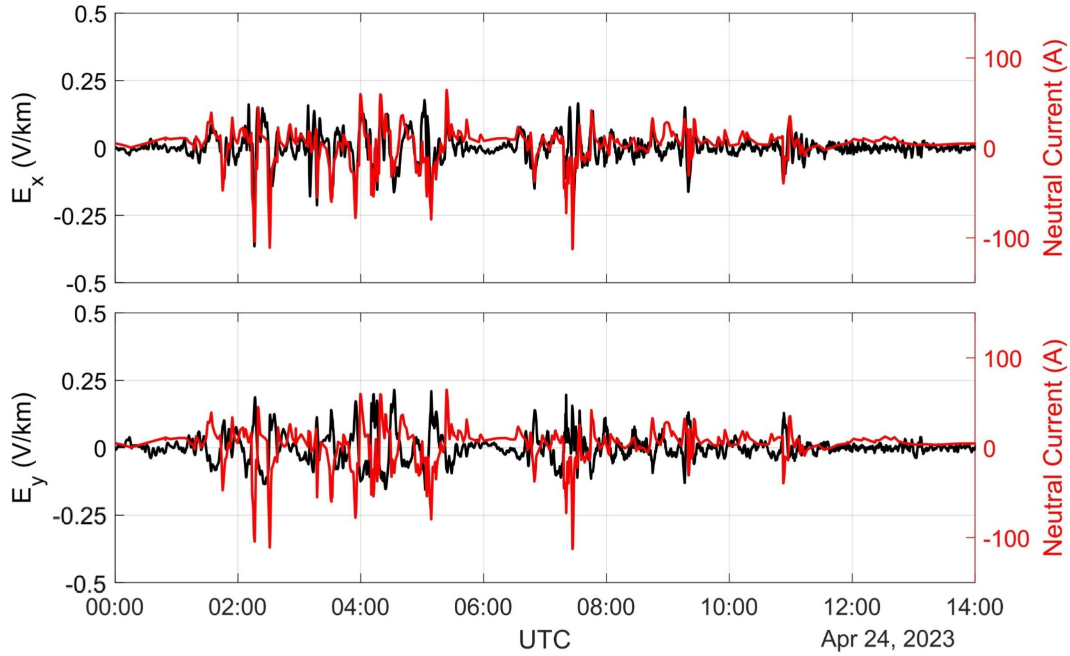

Figure 3. The geoelectric field northward, *x* (top), and eastward, *y* (bottom), components, estimated at MSTK magnetometer station. Overplotted in each panel is the TNG current in Amperes (red) from a nearby substation in central Alberta on 24-04-2023.

there could certainly be variations of the GIC in the power line that feeds into the substations that are a consequence of small-scale geoelectric fields integrated over the length of the power line but which are not seen in the data at MSTK approximately 35 km from the substation. However, it is appealing to suggest that the fact that the largest waveform discrepancies between the TNG GIC and the derived electric field occur at the times of the largest magnitude means that the smaller scale structures are also more important and perhaps also more prevalent at the times of maximum amplitude. There could be physical reasons in relation to how electric currents are closed in the ionosphere during space storms though this is beyond the scope of this study and is not investigated here.

It may be surprising to observe that at MSTK, E_x and E_y are well-correlated in anti-phase relationships and with a linear correlation coefficient of $R = -0.82$ over the entire time interval. This suggests that the Earth induction process is responsible for polarizing the electric field, despite a largely unpolarized driving magnetic field. This is important because the x and y components of the E-field contribute emf to portions of the power line which are aligned in the x and y directions, respectively. Therefore the topology of the power line in relation to the electric field polarization and their relative waveform and phase will have an important impact on whether the overall resulting induced currents for each electric field component are additive or cancel each other out. Combinations of network configuration and preferred geoelectric field polarization are investigated in more detail in Section 5 concerning identifying and assessing GIC risk.

4.2. Electric Field Characteristics: Correlation and Waveform Correspondence Between E_x and E_y

The three-dimensional conductivity structure of the Earth can modify the amplitude, phase, and polarization characteristics of the induced electric field driven by GMD (e.g., Love et al., 2022). The correlation of the horizontal geoelectric field components with each other occurs as result of the spatial structure of the underlying geology, and does not typically occur for an impedance produced by only a 1-D conductivity, and which has typically been used in prior historical studies. In the purely 1-D case, the diagonal elements of Z are zero. However, measured MT data always have non-zero diagonal components due to for example multi-dimensional conductivity effects, or noise. Specifically, if the diagonal Z_{xx} and Z_{yy} tensor elements are much smaller than the off-diagonal elements in all reference frames, then the tensor is effectively quasi-1-D. In 1-D, E_x and E_y are driven solely a result of B_x and B_y, respectively, in the frequency domain. Thus if the driving B_x and B_y are uncorrelated, then E_x and E_y should similarly also be uncorrelated. However, for a 3-D subsurface conductivity structure, E_x

10.1029/2025SW004544

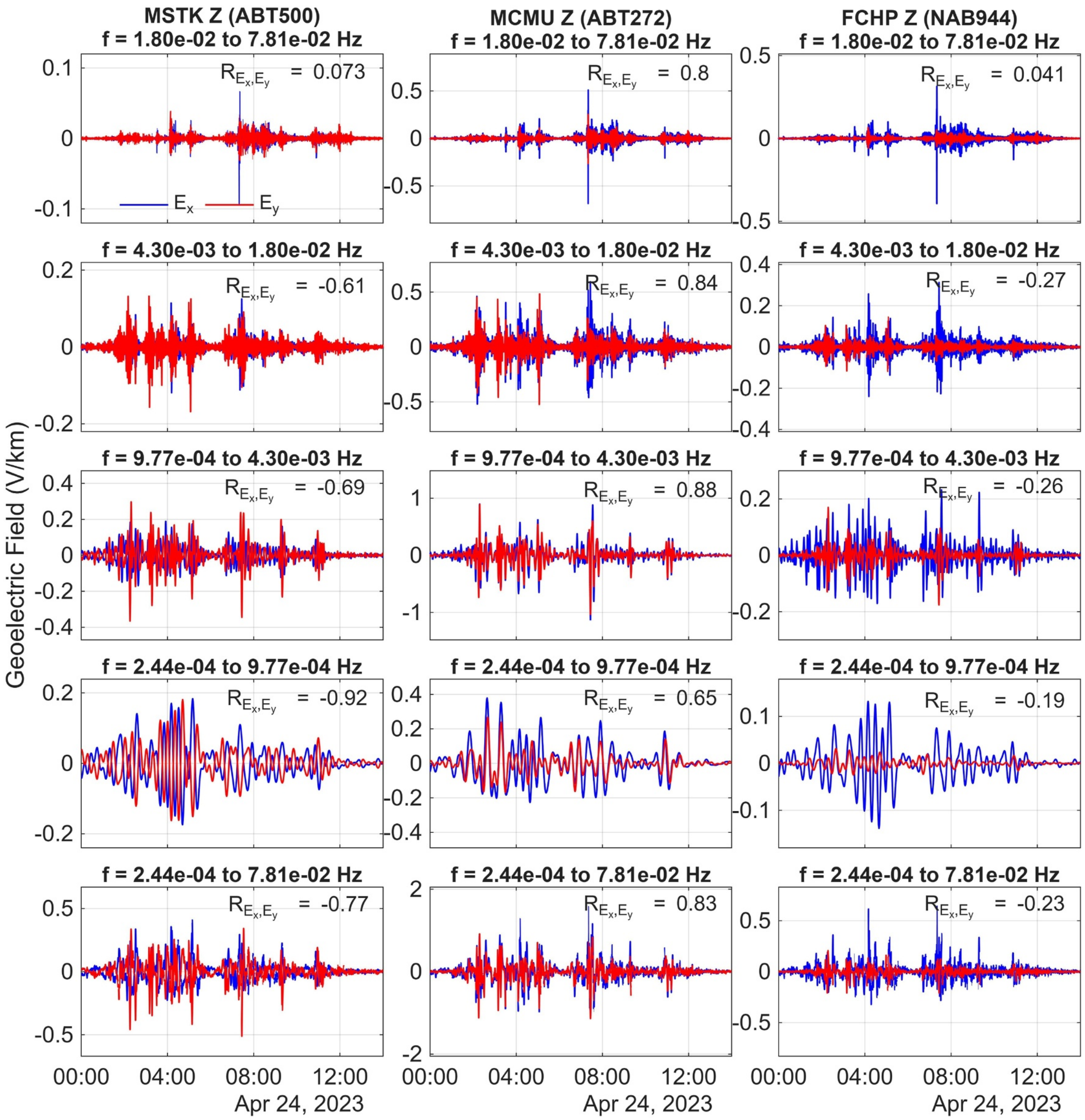

Figure 4. Geoelectric field, E_x (blue) and E_y (red), calculated using measured magnetic perturbations at MSTK station on 24-04-2023 but using three different surface impedances Z derived from MT measurements from sites near the magnetometers at MSTK (left column), at MCMU (center column), and at FCHP (right column). The linear correlation is calculated between the E_x and E_y components in each sub-frequency band, with decreasing frequency bands from the top to fourth row, and the total shared frequency range in the bottom row. See text for more details.

and E_y are each a product of both B_x and B_y, and therefore can share spectral elements. Depending on the specific characteristics of the frequency dependent impedance tensor, this can result in a correlation between E_x and E_y for specific underlying geologies.

We further demonstrate the importance of the measured MT surface impedance, which incorporates 3-D conductivity structure effects for controlling the characteristics of the resulting electric fields, by using a consistent driving ground magnetic field disturbance and accounting for the location-dependent impedance characteristics by using different Z. In Figure 4, the x and y geoelectric field components, blue and red respectively, are calculated using the same driving ground geomagnetic field perturbations from the MSTK magnetometer station on 24-04-2023. However, in Figure 4 we use three different impedance tensors taken from three distinct locations at MSTK (ABT500), MCMU (ABT272), and FCHP (NAB944) in deriving the resulting electric field. We further calculate the linear correlation between E_x and E_y across sub-frequency bands to investigate the electric field correlation described above in each case. While the frequency range of each MT survey varies slightly (ABT500: 0.0763–132.8120 mHz; ABT272: 0.1007–78.1250 mHz; NAB944: 0.1343–132.8120 mHz), the frequency bands for this comparison are chosen based on the shared frequency range of the MT surveys at all three locations (0.2441–78.1250 mHz). We then further separate the total frequency range into four smaller bands: 17.9688–78.1250 mHz, 4.2969–17.9688 mHz, 0.9766–4.2969 mHz, and 0.1343–0.9766 mHz. In Figure 4, the sub-frequency bands are arranged from higher frequencies in the top row to lower frequencies in the fourth row, and with the whole frequency spectrum shown in the bottom row.

The left column of Figure 4 shows the electric fields derived when MSTK Z is combined with MSTK magnetic fields. In this case, E_x and E_y are correlated in anti-phase as previously shown in the time domain in Figure 3. A high anti-phase correlation is maintained for all of the subsets of frequency ranges, particularly at low frequencies (0.317–1.32 mHz) with a correlation coefficient reaching $R = -0.88$. In contrast, the E_x and E_y resulting from combining MCMU Z with MSTK magnetic fields (middle column) are strongly correlated in phase ($R < 0$) in all frequency bands. Finally, the E_x and E_y resulting from combining FCHP Z and MSTK magnetic field (right column) are very weakly anti-correlated and strongly polarized in the x direction. The result derived by using FCHP Z would be that which is expected for a 1-D Earth, or from a driving signal which has a strongly polarized magnetic field in the east–west direction. For the whole frequency range, the correlation between E_x and E_y using the impedances from sites close to the MSTK, MCMU and FCHP stations are $R = -0.82$, 0.82, and 0.22, respectively. Again, since the same ground MSTK magnetic field is convolved with the Z transfer function from each different location in each case, the difference between the correlation values is directly attributable to the characteristics of the respective ground impedance tensor at the three different locations. Thus, it is clear that uncorrelated magnetic field components (i.e., randomly oriented B variations) can induce correlated electric field components (i.e., polarized E fields) due to interactions with the solid Earth, depending specifically on the characteristics of the impedance tensor at that location.

The differences in the magnitudes of the geoelectric field components arising from variations in Z are also shown in Figure 4. The magnitudes of E_x and E_y are five and ten times greater, respectively, for MCMU Z than for MSTK Z. At lower frequencies, the geoelectric field calculated using MCMU Z and MSTK Z are comparable in magnitude in E_x and only two times greater in E_y when calculated using MCMU Z than with MSTK Z. The geoelectric field magnitude calculated using FCHP Z is also much larger in the x component, but remains relatively weak across all frequency bands in comparison to the geoelectric field calculated using MCMU Z.

4.3. Preferred Geoelectric Field Polarization Direction

The polarization direction of the induced geoelectric field at MSTK is also of interest and is plotted in Figure 5 for 24-04-2023. The polarization is represented as the E-field bearing in degrees E of N. This is the angle between geographic North and the direction of the horizontal geoelectric field measured positive toward east. In the top panel of Figure 5, a scatter plot of the E-field bearing as a function of time is plotted. The E-field bearing is on the y-axis and the magnitude of E is represented by the color bar. An interval of large geoelectric field magnitude, from 04:08 – 06:08 UT (shaded yellow), is determined to be approximately spanning one hour prior to, and one hour following, the maximum geoelectric field magnitude, 0.603 V/km, which was observed at 05:08 UT on 24-04-2023. A comparator two-hour relatively geomagnetically quiet interval was chosen to be from 18:00 – 20:00 UT (shaded purple). At MSTK, the geoelectric field bearing is strongly polarized primarily in the directions N135 E and N315 E. At the times of the largest geoelectric fields such as at 02:15 UT, 05:08 UT, and 07:27 UT the electric field is also largely polarized in this direction. In the bottom panels of Figure 5, a more visual representation of the statistical variations of the geoelectric field bearing is shown. In the bottom left panel, a polar histogram shows that for both times of minimum (purple) and maximum (yellow) geoelectric field, the majority of the electric field remains strongly polarized NW-SE for MSTK Z (ABT500). This is also illustrated by the hodograph on the bottom right of Figure 5 showing the magnitude of the geoelectric field during the period of maximum geoelectric field (yellow) and the quiet interval (purple). On this scale, the amplitudes during the quiet time interval are barely perceptible. Thus, at MSTK the induced geoelectric field is predominantly polarized in the northwest-southeast direction regardless of the direction of the GMD variations. This is in agreement with the polarization ellipses of the impedance in Figure 2, which are oriented NW-SE in both periods. As discussed in Section 3, this Z characteristic, and the resulting preferred polarization of E, can likely be attributed to large crustal conductors trending NE-SW that lie beneath the WCSB (Figure 1).

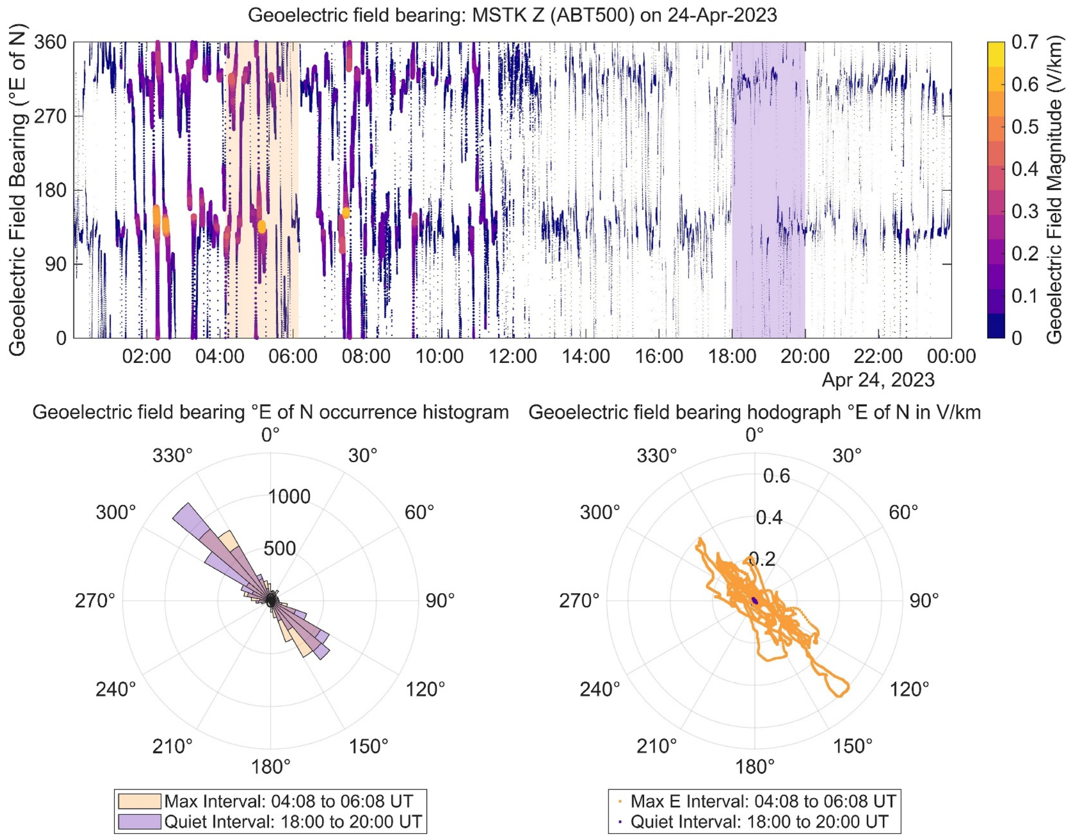

Figure 5. (top) A scatter plot of the geoelectric field bearing, in °E of N, using MSTK B and MSTK Z (ABT500) on 24-04-2023 is plotted with geoelectric field magnitude, in V/km, shown as the color of the line and represented by the color bar. Two selected intervals of large and quiet geoelectric field magnitude are shaded in yellow and purple, respectively. The magnitude and direction of the geoelectric field during large amplitude (yellow) and relative quiet (purple) intervals are plotted as a polar occurrence histogram (bottom left) and a hodograph (bottom right).

We repeat this analysis for MCMU Z (ABT272) using the same MSTK ground magnetic field perturbations as the driver of the E-field (Figure 6). When using MCMU Z, there is a preferred polarization of the geoelectric field in the NE-SW direction, with the preferred E-field bearing tending toward approximately N60 E and S60 W, in agreement with the impedance polarization ellipse at ABT272 in Figure 2. The resulting E-fields obtained using MCMU Z are preferentially polarized approximately perpendicular to the preferred polarization direction observed at MSTK. This preferred E-field polarization direction likely occurs due to the complex underlying geology including electrical anisotropy in the basement rocks beneath the thin WCSB (Liddell et al., 2016). This manifests as complex 3-D conductivity structures in northeast Alberta near MCMU at ABT272 (Figure 3). Finally, the E-field which results when using FCHP Z (NAB944) and MSTK magnetic field shown in Figure 7, has a polarization which is almost entirely east–west, with the preferred E-field bearing approximately 90° and 270°E of N. This is in agreement with the impedance polarization ellipse at NAB944 in Figure 2.

FCHP is located on the resistive rock of the Canadian Shield and deeper structures responsible for the polarization of the geoelectric field at MCMU may also influence the polarization at FCHP. Due to the more limited spatial resolution of MT data in northeastern Alberta, the 3-D conductivity model is much less constrained near FCHP and this makes it difficult to infer and use the trends of regional geophysical impedance structures to fully explain the geoelectric field polarization there. Nonetheless, previous geological mapping indicates that NAB944 lies along the N–S boundary between the Paleoproterozoic Taltson Magmatic Zone and Archaen Rae Craton (Chacko et al., 2000), which may provide a regional conductivity contrast that could be better mapped with greater MT coverage. In the Supporting Information S1 for this paper, we repeat the above analysis for another GMD event and show that the preferred polarization direction of the geoelectric field at these locations is not storm-dependent, and the resulting E-field polarization remains consistent with the explanation in terms of the characteristics of the

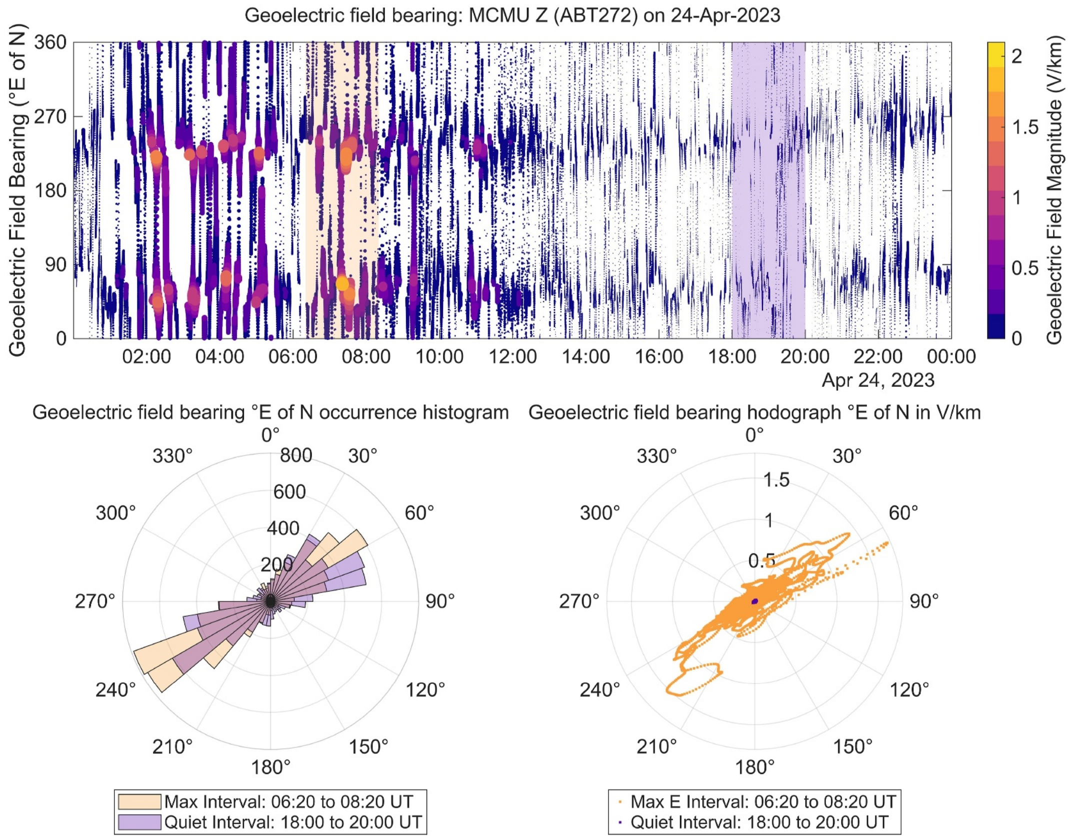

Figure 6. (top) A scatter plot of the geoelectric field bearing, in °E of N, using MSTK B and MCMU Z (ABT272) on 24-04-2023 is plotted with geoelectric field magnitude, in V/km, shown as the color of the line and represented by the color bar. Two selected intervals of large and quiet geoelectric field magnitude are shaded in yellow and purple, respectively. The magnitude and direction of the geoelectric field during large amplitude (yellow) and relative quiet (purple) intervals are plotted as a polar occurrence histogram (bottom left) and a hodograph (bottom right).

ground impedance tensor shown here (Figures S2–S4 in Supporting Information S1). For completeness of comparison, we also repeat this analysis using local driving magnetic field perturbations and impedance tensor (Figures S5 and S6 in Supporting Information S1).

In an attempt to validate the hypothesis that the E-field polarization at FCHP is a result of sub-surface impedance structures, the geoelectric field was calculated using the same MSTK magnetic field perturbations as above, but instead of using the full-tensor MT impedance measurement, a synthetic impedance tensor for a constant conductance 100 Ωm half-space is used. In this case, where the impedance tensor represents a single-layer geology, the polarization of the geoelectric field will be determined solely by the polarization of the horizontal magnetic components because the diagonal components of the tensor are identically zero (Equation 1). As seen in results shown in Figure 8, the horizontal magnetic field does not have a clear preferential polarization, and therefore there is no preferential polarization for the geoelectric field. As a result it seems likely that the E-field polarization seen in Figure 7 is also due to the underlying geological structure.

5. The Importance of Grid Topology: GIC Risk for Different Power Line Orientations

5.1. Simplified Model Example

The preferred polarization of the geoelectric field which can result at each location due to the local Z arising from the underlying geology can be used to help determine the GIC risk on a given electric power network based on its topology and relative orientation of line elements in the power grid and any preferred E-field polarization direction. We have developed a truth table based in the results of a simplified model to demonstrate this point. The truth table in Figure 9 shows two geoelectric field scenarios and two simplified power line geometries, for a case

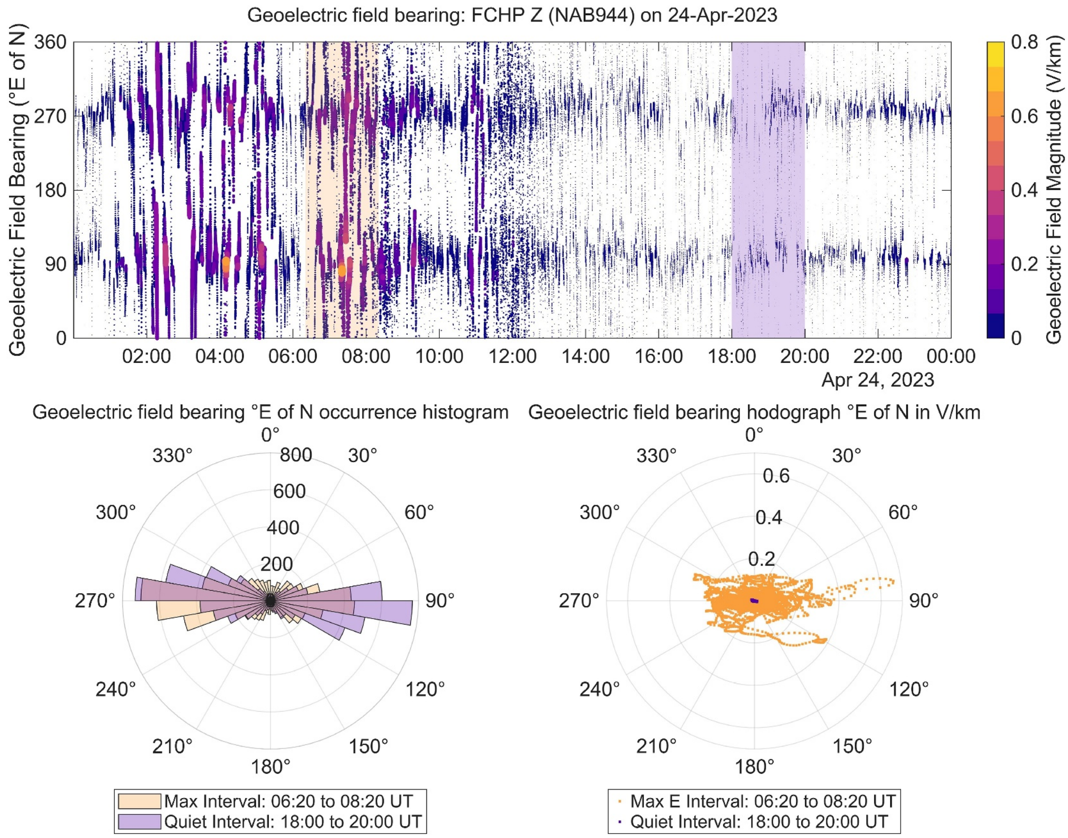

Figure 7. (top) A scatter plot of the geoelectric field bearing, in °E of N, using MSTK B and FCHP Z (NAB944) on 24-04-2023 is plotted with geoelectric field magnitude, in V/km, shown as the color of the line and represented by the color bar. Two selected intervals of large and quiet geoelectric field magnitude are shaded in yellow and purple, respectively. The magnitude and direction of the geoelectric field during large amplitude (yellow) and relative quiet (purple) intervals are plotted as a polar occurrence histogram (bottom left) and a hodograph (bottom right).

where the E_x and E_y components of the electric field are highly correlated and either in-phase or in anti-phase. Each power line configuration consists of two equal-length and perpendicular segments of a single power line connected at either end to a substation representing the connections to the rest of the power grid. The two grid geometries are mirror images of each other: both have a N–S line segment which is connected to the network on its north end, labeled A, and an E–W line segment connected to the network on either its west end, labeled B (top row), or east end, labeled C (bottom row). In the left column, E_x and E_y have the same amplitude and are in-phase ($\Delta\Phi = 0°$) like at MCMU, and in the right column, E_x and E_y are in anti-phase ($\Delta\Phi = 180°$), like at MSTK. Considering Section 4.1 when the electric field components are in phase (in anti-phase) the resulting emf along the line ΔV_{AB} is maximized (minimized). The reverse is true for the connection CA (bottom row).

We also consider the case with the same driving geoelectric field with in-phase and anti-phase polarizations with a similarly simple power line geometry, but with an additional grounding point inserted at the intersection of the N–S and E–W line segments (Figure 10). The grounding point is represented as a circle for the truth table in Figure 10. In this case, when the geoelectric field is polarized NE-SW and the power lines enter the substation from the north and west (top left) or when the geoelectric field is polarized NW-SE and the power lines enter the substation from the north and east (top right), minimum GIC flows to ground at the substation. In these cases, GIC flows through the substation bus either from the E–W power line to the NS power line or vice versa. Since the ground is not a perfect resistor, some GIC will flow to ground, however, most GIC flows through the substation assuming the power lines segments are the same voltage level and have the same line resistance (Boteler et al., 1994). Using Kirchhoff's Law, this can be represented as $|I_{TNG}| = |I_2 - I_1|$, where I_{TNG} is the current flowing through the transformer neutral-to-ground connection, and I_1 and I_2 are the induced currents in the N–S line segment and E–W line segments, respectively. In the first case (top row of Figure 10), the GIC combines

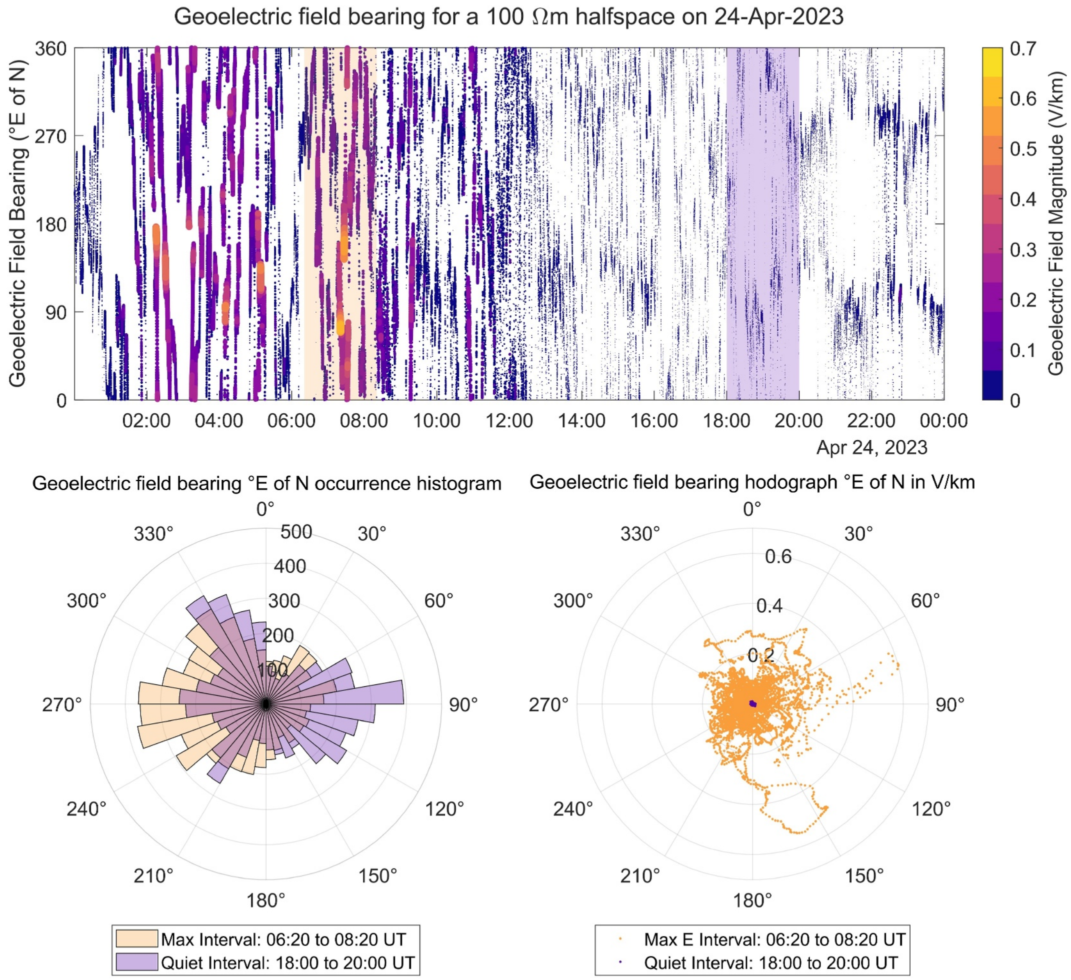

Figure 8. (top) A scatter plot of the geoelectric field bearing, in °E of N, using B measured at MSTK magnetometer station and Z for a 100 Ωm halfspace model on 24-04-2023 is plotted with geoelectric field magnitude, in V/km, shown as the color of the line and represented by the color bar. Two selected intervals of large and quiet geoelectric field magnitude are shaded in yellow and purple, respectively. The magnitude and direction of the geoelectric field during large amplitude (yellow) and relative quiet (purple) intervals are plotted as a polar occurrence histogram (bottom left) and a hodograph (bottom right).

destructively at the node. In contrast, when the geoelectric field is polarized exactly NE-SW and the power lines enter the substation from the north and east (bottom left) or when the geoelectric field is polarized exactly NW-SE and the power lines enter the substation from the north and west (bottom right), maximum GIC will flow to ground at the substation. In the latter case, I_1 and I_2 combine constructively such that $I_{TNG} = I_1 + I_2$. Again, this assumes the two power line segments have the same voltage level and line resistance, but will generally hold even if they differ somewhat. Overall in this case, the induced current in each power line segment adds constructively at the substation node resulting in a maximum GIC flowing through the substation to ground. This relatively straightforward exercise demonstrates the importance of the impedance and electric field polarization at the nodal level within the overall electrical power network. Although a network-wide DC model is required to understand the dynamics of the system as a whole, the analysis presented here emphasizes the combinatorial effect of the geoelectric field and power grid topology on the resulting GIC. It also implies that understanding the characteristics of any preferred geoelectric field polarization at a given location due to the 3-D subsurface conductivity structures can inform decisions about initial designs and perhaps also inform operation procedures which could be implemented within a specific portion of the grid during times of large GMD to limit the magnitude of GIC flowing through identified high-risk elements of the network.

10.1029/2025SW004544

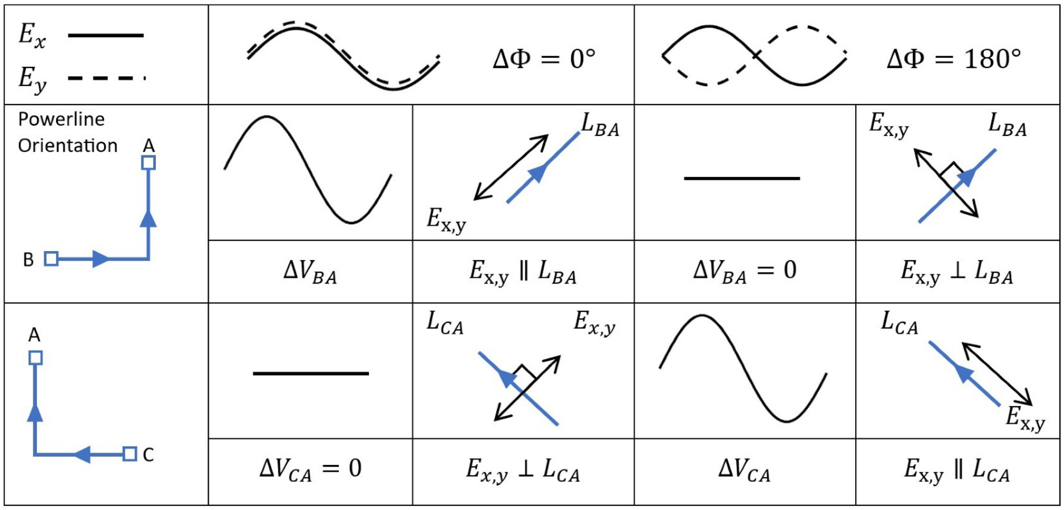

Figure 9. A truth table showing the voltage along the total line segment with orientation of either BA (second row) or CA (third row) in combination with a driving geoelectric field with polarization direction either NE-SW/E_x and E_y in-phase ($\Delta\Phi = 0$; left column) or NW-SE direction/E_x and E_y anti-phase ($\Delta\Phi = 180$; right column). The blue squares represent connections into the rest of the network. See text for more details.

5.2. Example From the Alberta Power Grid: Two Real-World Network Configurations

Two substations in the Alberta 500 kV network present an interesting real-world case study of the influence of network topology discussed in the previous section (Figure 11). Substation 1 is connected to a 500 kV transmission line from the east and a 240 kV line from the south, while Substation 2 is connected to two 500 kV transmission lines from the west and north. Substation 2 also has two connected 240 kV lines, however, their relative short lengths, approximately 15 km, limits the induced emf making them unlikely to carry significant GIC which would flow to ground here. The two substations are directly linked by one 500 kV transmission line (see Figure 11).

Based on the results from Section 5.1, we expect that this real-world example would show the result that, regardless of the geoelectric field orientation, the neutral-to-ground GIC from Substations 1 and 2 will always be

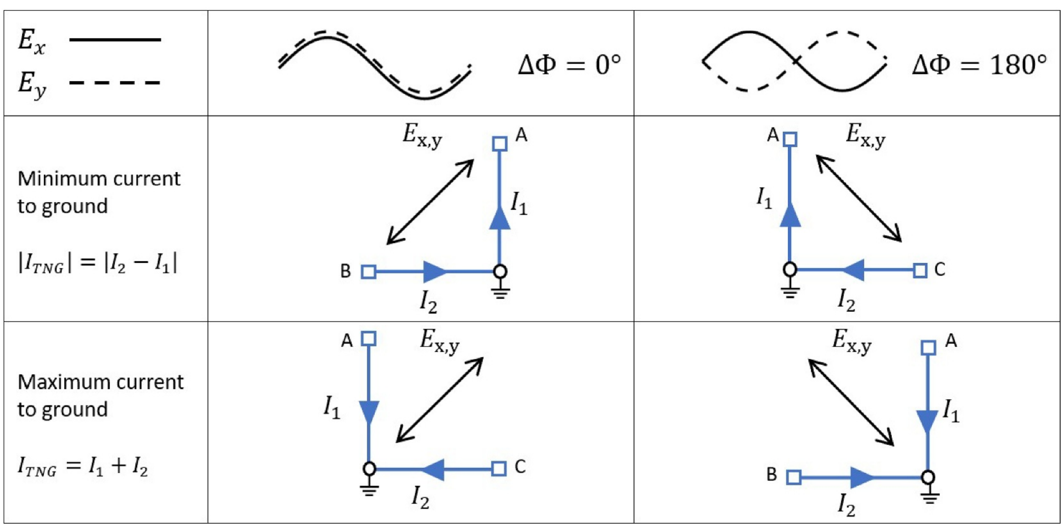

Figure 10. A second truth table shows cases for maximum and minimum induced current to ground as a result of combinations of geoelectric field polarization and power line geometry and configuration at a nodal level. Substations with paths for current to go to ground are symbolized as circles. The blue squares represent connections into the rest of the network.

10.1029/2025SW004544

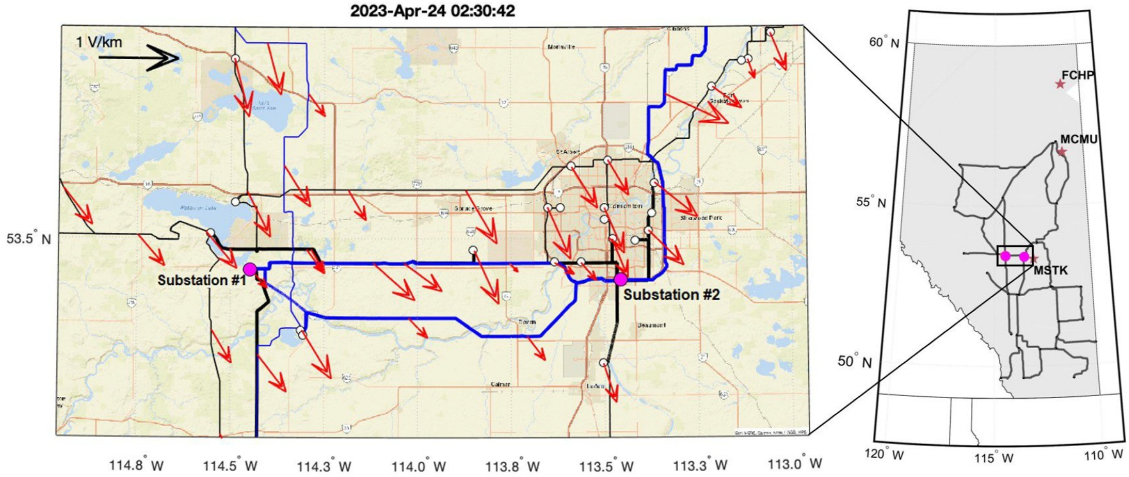

Figure 11. (left) Map of the transmission network in central Alberta along with calculated geoelectric field at query points along transmission lines at 02:30:42 UT on 24-04-2023. White dots are substations. Blue and black lines are 500 and 240 kV transmission lines, respectively. Thicker lines show transmission lines entering Substation 1 and/or Substation 2. (right) Map of Alberta including the Alberta Interconnected Electric System (AIES) (dark gray) and approximate locations of two Hall effect sensors (pink circles) shown in a zoomed-in figure in the left panel.

opposite to one another, resulting in a negative correlation between TNG GIC time series at these two substations (see Figure 12). If the geoelectric field is oriented north, then current will preferentially flow out of Substation 2 and into Substation 1 because the emf on the Substation 2 transmission lines will be negative (i.e., away from the substation) while it will be positive at Substation 1 (i.e., toward the substation). The same is true of any other orientation. This is indeed what is observed in the real-world Hall sensor neutral-to-ground GIC measurements at these two substations. An example is shown from 24-04-2023, between 01:00 and 09:00 UT in Figure 12.

Based on the network topology, we would also expect this effect to be maximized when the geoelectric field is oriented approximately northwest or southeast as per Figure 10. If we run a GIC simulation model of the >240 kV Alberta Interconnected Electric System (AIES) as described in Cordell et al. (2024) using a uniform 1 V/km unidirectional geoelectric field oriented at variable directions in 5° increments, we see that Substation 2 has a peak GIC of +37 A (into the substation) when the geoelectric field is oriented 145°E of N (i.e., southeast) and a trough of −37 A (out of the substation) when the geoelectric field is oriented 325°E of N (i.e., northwest) (Figure 13). Substation 1 is inversely correlated with a peak and trough of ±43 A when the geoelectric field is oriented 300°E of N (i.e., northwest) and 120°E of N (i.e., southeast), respectively. As shown in Figure 13, the correlation and

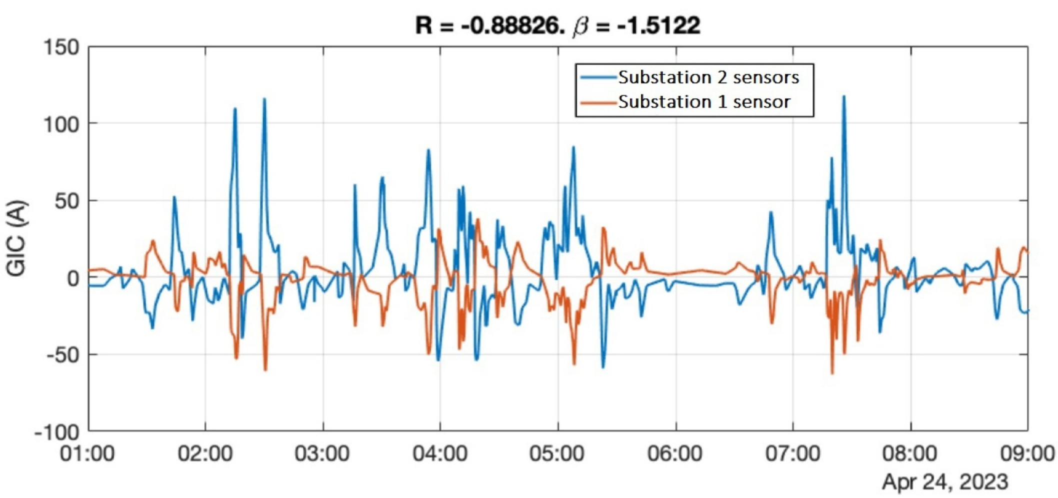

Figure 12. Plot of GIC measured at Substations 1 and 2 measured using Hall effect sensors on the transformer neutral-to-ground connections at these substations.

10.1029/2025SW004544

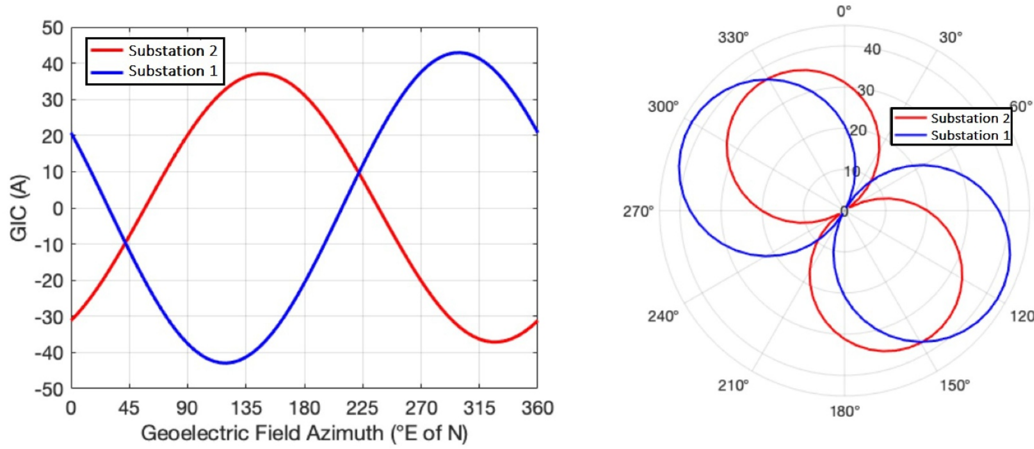

Figure 13. Plots of TNG GIC at Substations 1 and 2 using a 1 V/km uniform geoelectric field rotated through 360° in 5° increments. Left panel shows GIC values as a function of azimuth. Right panel shows a polar plot of GIC magnitude as a function of azimuth.

magnitudes are not identical because of the influence of current leakage onto other transmission lines and lower voltage levels. In particular, both Substation 1 and 2 have autotransformer banks which will allow some current leakage directly to the 240 kV bus. The negative correlations and matching magnitudes as a function of uniform and unidirectional geoelectric field polarization angle in azimuth highlights the role of network topology in determining the magnitude of the TNG GIC, independent of any effects arising additionally from non-uniform spatial geoelectric field variations and which will of course vary from event to event.

What is particularly interesting is that the effects of this network topology is combined with the electric field polarization effects of the ground impedance characteristics of the region to introduce especially large TNG GICs. As shown in Figures 2 and 5, the geoelectric field tends to be preferentially polarized in the northwest-southeast direction in central Alberta due to the influence of the ground conductivity. Figure 5 shows that largest geoelectric fields will also tend to be oriented in that direction. This happens to be the same direction which will maximize GICs at these substations as per Figure 9. As noted in Cordell et al. (2024), Substations 1 and 2 have the largest modeled and measured GIC in the province. This is likely explained by the combination of the local ground conductivity aligning the preferred electric field direction such that the overall TNG GIC the E-fields produce is maximized due to the specific orientations and connections in the network topology at this location.

To highlight this, we calculated the geoelectric field across the province during the GMD on 24-04-2023, using the data-space method described in Cordell et al. (2021, 2024), in which the geoelectric field is calculated at each MT site. We specifically focus on the time interval from 01:00 to 09:00 UT, during the main phase of the storm. We computed the average magnitude and direction of the geoelectric field from across the entire province from the model at each time step. This is shown as a polar plot in Figure 14a. As can be seen, there are several large geoelectric field events reaching average magnitudes of ∼0.4 V/km. The largest geoelectric fields tend to be oriented southeast although there are two exceptions: during one particular interval the electric field azimuth varies between 60° and 120°E of N with magnitudes >0.3 V/km, and during another interval is oriented 30°E of N with a magnitude of ∼0.3 V/km. Figure 14b shows a histogram of the geoelectric field azimuth which makes it clearer that there is, in general, a northwest-southeast polarization of the average geoelectric field across the province, partially due to the effects of the ground impedance as described in Section 3.

Figures 14c and 14d show the measured GIC magnitude at Substation 2 and Substation 1, respectively. As can be seen, the GIC magnitude is greatest in the southeast direction. However, notably (and especially at Substation 2), the two large geoelectric field events, but which are not oriented southeast, have a very limited GIC response. For example, the large 0.3 V/km event with azimuth oriented at 30°E of N results in a GIC of <50 A, whereas a similar size geoelectric field oriented southeast results in a GIC of >100 A. This highlights that the largest GICs at these substations are strongly modulated by the geoelectric field direction. Even a large geoelectric field oriented northeast results in significantly less GIC than smaller electric field magnitudes oriented southeast. Conversely, a

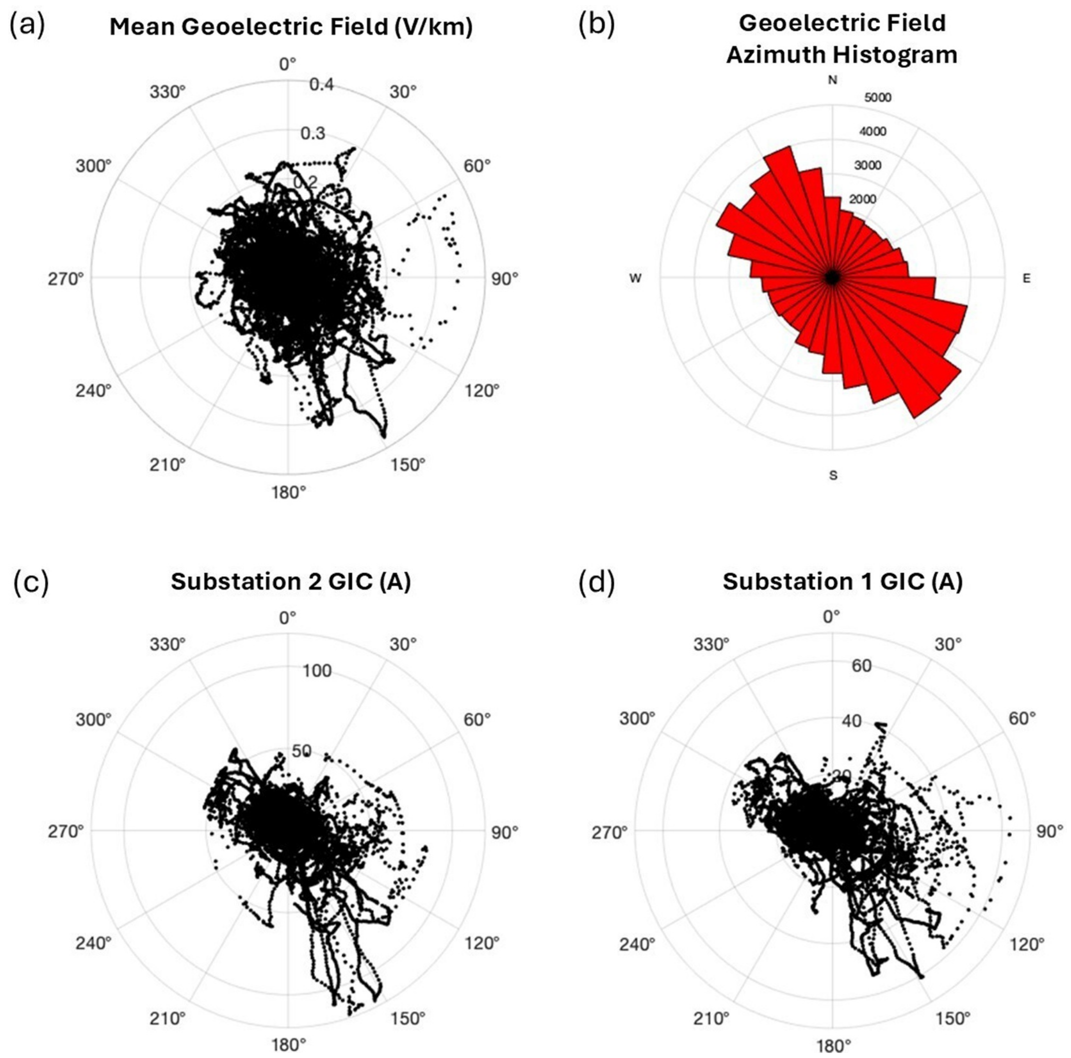

Figure 14. Plot of magnitude and direction of geoelectric field and GICs. Top left (a): Plot showing the average calculated geoelectric field magnitude and direction across the whole province. Top right (b): Plot showing histogram of geoelectric field azimuth. Bottom left (c) and right (d): Each dot plots the magnitude of the GIC with the geoelectric field azimuth for a single time sample. Data is taken from 01:00 to 09:00 UT on 24-04-2023. See text for details.

geoelectric field oriented southeast results in a larger GIC than might be expected given the geoelectric field magnitude alone.

As shown with this example, the combination of network topology and ground conductivity helps explain why the GICs at Substations 1 and 2 tend to be the largest in the 500 kV network (Cordell et al., 2024). Electric current is concentrated there due to the GIC being maximized when the geoelectric field is oriented northwest-southeast, and this preferred electric field polarization is observed more often than other polarization directions given the ground impedance effects of this region.

6. Conclusions and Future Work

This study examines the role of Earth's subsurface conductivity structure in determining the polarization and phase characteristics of the induced geoelectric field during space weather-driven geomagnetic disturbance events. Analyzing three 3-D impedance tensors across different geological regions in Alberta, Canada, we demonstrated that electric field phase and polarization characteristics can be strongly influenced by ground conductive effects. In particular, when considering the full 3-D conductivity description of the subsurface geology, specific conductivity structures can result in the horizontal E-field components, E_x and E_y, that exhibit

10.1029/2025SW004544

strong temporal and waveform correlation. Consequently, the geoelectric field responsible for driving currents through power system components at such locations may have preferred electric field polarization directions. We use the 3-D MT impedance tensor, and observed magnetic field measurements from MSTK CARISMA magnetometer stations to examine such behavior in the province of Alberta, Canada. We find that at MSTK E_x and E_y are well correlated and in anti-phase, and the geoelectric field has a preferred polarization direction of 35° and 315°E of N. At MCMU, E_x and E_y are in-phase and the geoelectric field has a preferred polarization direction of 60° and 240°E of N. Finally, at FCHP, E_x and E_y are very weakly correlated and the geoelectric field has a preferred polarization direction 90° and 270°E of N. While both MSTK and MCMU stations are located on geology within the WCSB, the polarization characteristics at these two locations are significantly different due the specific local geologies and the different relative proximity to different conductive structures. The typical 1-D Earth model for previous GIC studies does not account for these differences of these two locations.

We suggest that these specific regions where E_x and E_y are highly correlated (i.e., the polarization direction of the geoelectric field is predominantly in either the NW-SE or NE-SW quadrants) and that this can significantly impact the magnitude and distribution of GICs in power networks with specific directional topologies at such locations. We present a truth table which illustrates the effects of this and shows that for a simple power line geometry consisting of two perpendicular power line segments in either the right-handed, L_{BA}, or left-handed, L_{CA}, orientation, the total applied voltage in either geometry (V_{BA} or V_{CA}) depends on the polarization direction of the geoelectric field (see Figure 9). When a geoelectric field is highly polarized in the NE-SW (E_x and E_y in-phase) direction, V_{BA} is maximum and V_{CA} is zero, whereas if the geoelectric field is highly polarized in the NW-SE (E_x and E_y anti-phase) direction V_{BA} is zero and V_{CA} is maximum. Another simple power line geometry is considered by adding a grounding point at the intersection of the two perpendicular line segments (see Figure 10). This configuration demonstrates that the GIC flowing along each line segment, I_1 and I_2, can either combine constructively or destructively at the node, resulting in either significantly enhanced or significantly reduced GIC flowing to ground.

The modeled and measured GIC during the April 2023 geomagnetic storm was also investigated in a portion of the AIES in central Alberta. This is a region where the underlying geology results in a strongly polarized geoelectric field, with a preferred electric field polarization with a bearing approximately 35° and 315°E of N, and where E_x and E_y are in anti-phase. This real-world example on the Alberta power network shows that for locations where the geoelectric field is strongly polarized in a NW-SE direction, the resulting GIC flowing to ground at the substations is enhanced based on the local topological configuration of the 500 kV portions of the network there. Long high-voltage power lines are connected into substation 1 from the south and east, and connect into substation 2 from the west and north. The real-world power line orientation and connection at this location, in combination with the preferred polarization direction of the geoelectric field in this region in central Alberta, can be described by the scenario in the bottom right portion of Figure 10, that is one where the combination of the local grid topology and the preferred electric field polarization combine to maximize the GIC flowing to ground at the transformer.

These findings highlight the critical influence and importance of 3-D subsurface conductivity on generated geoelectric fields with a preferred polarization, and the resulting enhancement in geomagnetically induced currents in power networks with specific orientations and topologies. Incorporating 3-D Earth impedance effects into GIC risk assessments appears to be essential for accurately predicting the spatial distribution and magnitude of the induced currents in power networks with a specific configuration. Ultimately, such effects are likely crucial for aiding in the mitigation of the potential adverse impacts on power infrastructure arising from space weather events. We can attempt to identify specific elements of any given power network that may be more susceptible to GIC impacts and therefore further inform future GIC risk mitigation strategies in the province of Alberta and elsewhere.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Magnetic field data for the Ministik Lake (MSTK), Fort McMurray (MCMU) and Fort Chipewyan (FCHP) magnetic station was provided by CARISMA and is freely available at CARISMA Data (2024). The presented magnetometer, impedance and Hall probe data are hosted on Zenodo at Parry (2025). Data supporting this research, including specific information concerning substation geographic locations and names, were made available by request from AltaLink L.P. and deemed sensitive information under an NDA, and are not accessible to the public or research community. Researchers seeking information can contact AltaLink at 403-267-3400.

Acknowledgments

We acknowledge the National Science and Engineering Research Council of Canada (NSERC) for support through Alliance Advantage and Discovery Grants to IRM and MJU, and a Postdoctoral Fellowship to DC. This research has been made possible via an NSERC Alliance Grant with sponsorship from AltaLink Management Ltd and ATCO Electric, and in-kind contribution from the Alberta Electric System Operator. We would like to thank Martyn Unsworth for providing MT data. CARISMA is operated by the University of Alberta and funded by the Canadian Space Agency.

References

Bedrosian, P. A., & Love, J. J. (2015). Mapping geoelectric fields during magnetic storms: Synthetic analysis of empirical United States impedances. *Geophysical Research Letters*, *42*(23), 10–160. https://doi.org/10.1002/2015gl066636

Berdichevsky, M. N., & Dmitriev, V. I. (2008). *Models and methods of magnetotellurics*. Springer Science and Business Media.

Bolduc, L. (2002). GIC observations and studies in the Hydro-Québec power system. *Journal of Atmospheric and Solar-Terrestrial Physics*, *64*(16), 1793–1802. https://doi.org/10.1016/s1364-6826(02)00128-1

Boteler, D. H. (1994). Geomagnetically induced currents: Present knowledge and future research. *IEEE Transactions on Power Delivery*, *9*(1), 50–58. https://doi.org/10.1109/61.277679

Boteler, D. H., Bui-Van, Q., & Lemay, J. (1994). Directional sensitivity to geomagnetically induced currents of the Hydro-Quebec 735 kV power system. *IEEE Transactions on Power Delivery*, *9*(4), 1963–1971. https://doi.org/10.1109/61.329528

Boteler, D. H., & Pirjola, R. J. (2017). Modeling geomagnetically induced currents. *Space Weather*, *15*(1), 258–276. https://doi.org/10.1002/2016sw001499

Boteler, D. H., Pirjola, R. J., & Marti, L. (2019). Analytic calculation of geoelectric fields due to geomagnetic disturbances: A test case. *IEEE Access*, *7*, 147029–147037. https://doi.org/10.1109/access.2019.2945530

CARISMA Data. (2024). CARISMA data [Dataset]. Retrieved from https://carisma.ca/

Chacko, T., De, S. K., Creaser, R. A., & Muehlenbachs, K. (2000). Tectonic setting of the Taltson magmatic zone at 1.9 2.0 ga: A Granitoid-based perspective. *Canadian Journal of Earth Sciences*, *37*(11), 1597–1609. https://doi.org/10.1139/cjes-37-11-1597

Cordell, D., Mann, I. R., Parry, H., Unsworth, M. J., Cui, R., Clark, C., et al. (2024). Modeling geomagnetically induced currents in the Alberta power network: Comparison and validation using hall probe measurements during a magnetic storm. *Space Weather*, *22*(4), e2023SW003813. https://doi.org/10.1029/2023sw003813

Cordell, D., & Unsworth, M. (2025). The influence of the geoelectric coast effect on geomagnetically induced currents. *IEEE Transactions on Power Delivery*, *40*(3), 1–11. https://doi.org/10.1109/tpwrd.2025.3544488

Cordell, D., Unsworth, M. J., Lee, B., Hanneson, C., Milling, D. K., & Mann, I. R. (2021). Estimating the geoelectric field and electric power transmission line voltage during a geomagnetic storm in Alberta, Canada using measured magnetotelluric impedance data: The influence of three-dimensional electrical structures in the lithosphere. *Space Weather*, *19*(10), e2021SW002803. https://doi.org/10.1029/2021sw002803

Cuttler, S. W., Love, J. J., & Swidinsky, A. (2018). Geoelectric hazard assessment: The differences of geoelectric responses during magnetic storms within common physiographic zones. *Earth Planets and Space*, *70*(1), 35. https://doi.org/10.1186/s40623-018-0807-7

Hanneson, C., & Unsworth, M. J. (2023). Regional-scale resistivity structure of the middle and lower crust and uppermost mantle beneath the southeastern Canadian cordillera and insights into its causes. *Geophysical Journal International*, *234*(3), 2032–2052. https://doi.org/10.1093/gji/ggad183

Jones, M.-F., Castonguay, M., Nasr, M., Ogilvie, J., Arp, P. A., & Bhatti, J. (2014). Modeling hydrothermal regimes and potential impacts of climate change on permafrost within the south Mackenzie plain, Northwest Territories, Canada. *Écoscience*, *21*(1), 21–33. https://doi.org/10.2980/21-1-3663

Kelbert, A. (2019). The role of global/regional Earth conductivity models in natural geomagnetic hazard mitigation. *Surveys in Geophysics*, *41*(1), 115–166. https://doi.org/10.1007/s10712-019-09579-z

Liddell, M., Unsworth, M., & Pek, J. (2016). Magnetotelluric imaging of anisotropic crust near Fort McMurray, Alberta: Implications for engineered geothermal system development. *Geophysical Journal International*, *205*(3), 1365–1381. https://doi.org/10.1093/gji/ggw089

Love, J. J., Lucas, G. M., Rigler, E. J., Murphy, B. S., Kelbert, A., & Bedrosian, P. A. (2022). Mapping a magnetic superstorm: March 1989 geoelectric hazards and impacts on United States power systems. *Space Weather*, *20*(5), e2021SW003030. https://doi.org/10.1029/2021sw003030

Malone-Leigh, J., Campanyà, J., Gallagher, P. T., Neukirch, M., Hogg, C., & Hodgson, J. (2023). Nowcasting geoelectric fields in Ireland using magnetotelluric transfer functions. *Journal of Space Weather and Space Climate*, *13*, 6. https://doi.org/10.1051/swsc/2023004

Mann, I. R., Milling, D. K., Rae, I. J., Ozeke, L. G., Kale, A., Kale, Z. C., et al. (2008). The upgraded CARISMA magnetometer array in the THEMIS era. *Space Science Reviews*, *141*(1–4), 413–451. https://doi.org/10.1007/s11214-008-9457-6

Marshall, R., Wang, L., Paskos, G., Olivares-Pulido, G., Van Der Walt, T., Ong, C., et al. (2019). Modeling geomagnetically induced currents in Australian power networks using different conductivity models. *Space Weather*, *17*(5), 727–756. https://doi.org/10.1029/2018sw002047

McKay, A., & Whaler, K. (2006). The electric field in northern England and southern Scotland: Implications for geomagnetically induced currents. *Geophysical Journal International*, *167*(2), 613–625. https://doi.org/10.1111/j.1365-246x.2006.03128.x

Molinski, T. S. (2002). Why utilities respect geomagnetically induced currents. *Journal of Atmospheric and Solar-Terrestrial Physics*, *64*(16), 1765–1778. https://doi.org/10.1016/s1364-6826(02)00126-8

Murphy, B. S., Lucas, G. M., Love, J. J., Kelbert, A., Bedrosian, P. A., & Rigler, E. J. (2021). Magnetotelluric sampling and geoelectric hazard estimation: Are national-scale surveys sufficient? *Space Weather*, *19*(7), e2020SW002693. https://doi.org/10.1029/2020sw002693

NERC. (2020). *Transmission system planned performance for geomagnetic disturbance events*. Tech. Rep. No. TPL-007-4. North American Electric Reliability Corporation.

Nieuwenhuis, G., Unsworth, M. J., Pana, D., Craven, J., & Bertrand, E. (2014). Three-dimensional resistivity structure of southern Alberta, Canada: Implications for precambrian tectonics. *Geophysical Journal International*, *197*(2), 838–859. https://doi.org/10.1093/gji/ggu068

Parry, H. (2025). Geomagnetic field, impedance tensor and Hall probe GIC data from 2024-04-23 in AB, Canada [Dataset]. *Zenodo*. https://doi.org/10.5281/zenodo.16748607

Parry, H., Mann, I., Kale, A., Milling, D., Clark, C., Cui, R., et al. (2024). Using a differential magnetometer technique to measure geomagnetically induced currents: An augmented approach. *Space Weather*, *22*(7), e2024SW003894. https://doi.org/10.1029/2024sw003894

Pirjola, R. (2000). Geomagnetically induced currents during magnetic storms. *IEEE Transactions on Plasma Science*, *28*(6), 1867–1873. https://doi.org/10.1109/27.902215

Pirjola, R. (2002). Review on the calculation of surface electric and magnetic fields and of geomagnetically induced currents in ground-based technological systems. *Surveys in Geophysics*, *23*(1), 71–90. https://doi.org/10.1023/a:1014816009303

Pulkkinen, A., Bernabeu, E., Eichner, J., Beggan, C., & Thomson, A. W. P. (2012). Generation of 100-year geomagnetically induced current scenarios. *Space Weather*, *10*(4). https://doi.org/10.1029/2011sw000750

Pulkkinen, A., Bernabeu, E., Thomson, A., Viljanen, A., Pirjola, R., Boteler, D., et al. (2017). Geomagnetically induced currents: Science, engineering, and applications readiness. *Space Weather*, *15*(7), 828–856. https://doi.org/10.1002/2016sw001501

Ross, G. M. (2002). Evolution of Precambrian continental lithosphere in western Canada: Results from Lithoprobe studies in Alberta and beyond. *Canadian Journal of Earth Sciences*, *39*(3), 413–437. https://doi.org/10.1139/e02-012

Ross, G. M., & Eaton, D. W. (2002). Proterozoic tectonic accretion and growth of western Laurentia: Results from Lithoprobe studies in northern Alberta. *Canadian Journal of Earth Sciences*, *39*(3), 313–329. https://doi.org/10.1139/e01-081

Schrijver, C. J., Dobbins, R., Murtagh, W., & Petrinec, S. M. (2014). Assessing the impact of space weather on the electric power grid based on insurance claims for industrial electrical equipment. *Space Weather*, *12*(7), 487–498. https://doi.org/10.1002/2014sw001066

Torta, J., Marcuello, A., Campanyà, J., Marsal, S., Queralt, P., & Ledo, J. (2017). Improving the modeling of geomagnetically induced currents in Spain. *Space Weather*, *15*(5), 691–703. https://doi.org/10.1002/2017sw001628

Trichtchenko, L., Nikitina, L., & Fernberg, P. (2015). Space weather hazard assessment for Alberta. *Geological Survey of Canada, Open File*, *7906*(156). https://doi.org/10.4095/296909

Türkoğlu, E., Unsworth, M., & Pana, D. (2009). Deep electrical structure of northern Alberta (Canada): Implications for diamond exploration. *Canadian Journal of Earth Sciences*, *46*(2), 139–154. https://doi.org/10.1139/e09-009

Unsworth, M. (2015). Magnetotelluric studies of lithospheric structure beneath western Canada: Insights into plate tectonics both past and present. *CSEG Recorder*, 28–32.

Utada, H., & Munekane, H. (2000). On galvanic distortion of regional 3-D MT impedances on galvanic distortion of regional three-dimensional magnetotelluric impedances. *Geophysical Journal International*, *140*(2), 385–398. https://doi.org/10.1046/j.1365-246x.2000.00014.x

Wang, E., & Unsworth, M. (2022). Three-dimensional crustal and upper-mantle resistivity structure of Alberta, Canada: Implications for Precambrian tectonics. *Geophysical Journal International*, *230*(3), 1679–1698. https://doi.org/10.1093/gji/ggac128

Zheng, K., Trichtchenko, L., Pirjola, R., & Liu, L.-G. (2013). Effects of geophysical parameters on GIC illustrated by benchmark network modeling. *IEEE Transactions on Power Delivery*, *28*(2), 1183–1191. https://doi.org/10.1109/tpwrd.2013.2249119